

Poverty in the United Republic of Tanzania

Figure 1

Measures of poverty with adjusted household expenditures		
	Headcount Index	Poverty Gap Index
Per capita expenditure	0.365	0.017
Square root economies of scale	0.023	0.001
Adult equivalence Scale	0.217	0.008

The headcount index gives us the ratio of people that would be considered poor given a certain poverty line (z-value; set here at \$1.90). The poverty gap index is also a ratio (between 1 and 0) expressing the depth of poverty; how far people are below the poverty line. Square root economies of scale and adult equivalence scales adjust the poverty line for a household based on household size and number of children respectively. This lowers the effect on the indexes from large households and houses with more children; which are likely to make less per capita. Children are less likely to earn and large households are more likely to contain children and other dependents like the sick or elderly. This would explain why applying square root economies of scale causes drastic changes to the headcount and poverty gap index, as seen in figure 1.

Figure 2

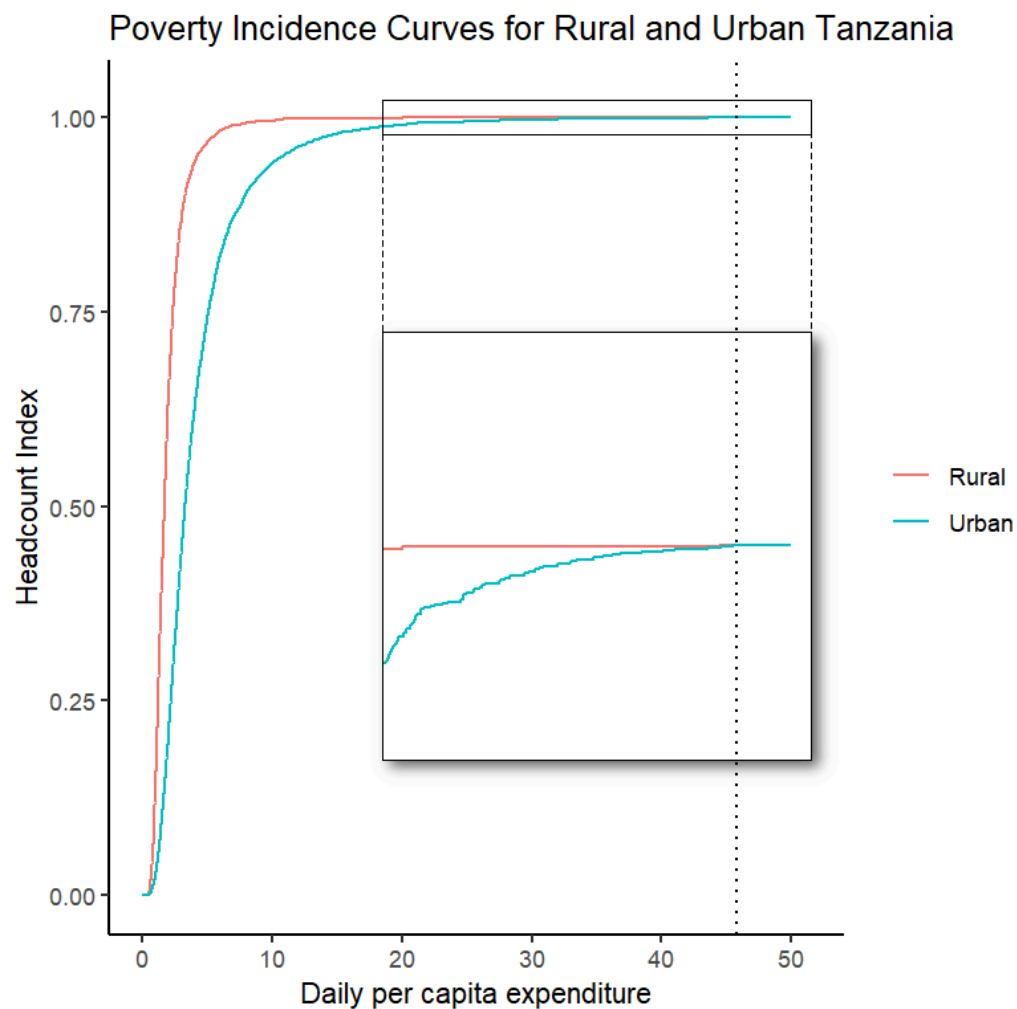


Figure 2 shows the head count index of Tanzanian rural and urban households using poverty lines set at different values for daily per capita expenditure. When only considering households with daily per capita expenditures under \$50, the poverty incidence curves intersect at 45.80, marked by the dotted line, where 100% of households would be considered poor. \$45.80 is the largest expenditure in this restricted dataset and belongs to an urban household. The largest rural per capita daily expenditure is \$44.45. There is no z-value where urban households experience more poverty, the two curves only intersect where the headcount index is at 0 and 1. This means we can make the robust comparison that rural households are unambiguously poorer than rural households in Tanzania.

Figure 3

T-Test Results Comparing Gender Differences in Household Outcomes in Tanzania

Parameters	Male mean	Female mean	Difference	95% CI ¹	t	df_error	p
Log per-capita expenditure	1.123	1.093	0.02938	-0.003018 0.06177	1.778	4,680	0.07550
Below poverty line	0.2708	0.2896	-0.01881	-0.03885 0.001240	-1.839	4,488	0.06595
Below poverty line adjusted for square-root economies of scale	0.0216 9	0.0423 0	-0.02061	-0.02899 -0.01224	-4.825	3,621	<0.00001

Poverty line set at \$1.90

¹ CI = Confidence Interval

The above results show that households with female heads earn slightly less and are slightly more likely to be below the poverty line, but a Welch two sample t-test determined the difference was marginally too small to be definitively statistically significant ($t < |2|$, $p > 0.05$).

The results also show that households with female heads are far more likely to be below the poverty line after adjusting household expenditure for square-root economies of scale and that this result is statistically significant. This, combined with data from figure 1, would suggest households with female heads are likely to be smaller. Given the majority of households in the dataset are male led, it would seem the expectation in Tanzania is that it is expected a man lead the household. A possible explanation for female households being smaller is that female led households are less likely to be headed by a married couple and are less likely to have kids.

Figure 4

Regression results for log household per-capita expenditure, daily

Characteristic	Beta	95% CI¹	p-value
Educated	1.5	1.0, 1.9	<0.001
Age of Household head	-0.03	-0.04, -0.02	<0.001
Married	-2.0	-2.5, -1.4	<0.001
Separated	-2.0	-2.6, -1.5	<0.001
Number of people in household	-0.14	-0.20, -0.08	<0.001
Share of household under 14	-5.9	-6.6, -5.2	<0.001

¹ CI = Confidence Interval

The characteristic that made the largest difference to the log of household per-capita expenditure was the share of the household under the age of 14 (Beta = -5.9), meaning households with larger shares of children are, on average, much poorer with lower daily per-capita expenditure. This likely reflects the financial and resource strain of supporting more dependents, we also see a small effect from simply having larger households

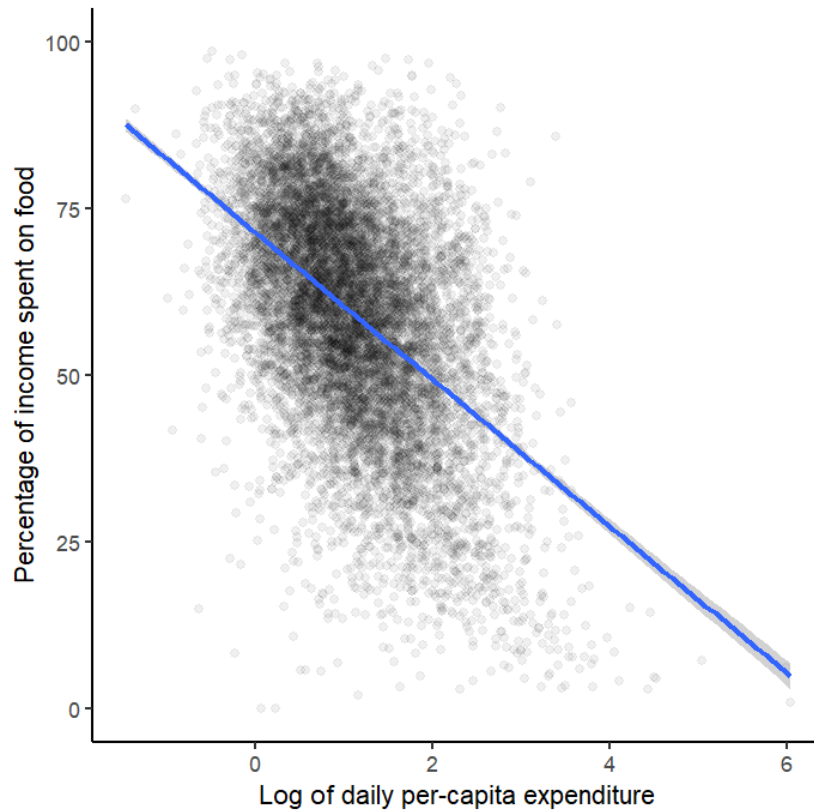
Both households with married/cohabiting heads and households with separated/widowed heads were expected to have significantly lower log daily per-capita expenditure when compared to households where the head was never married. These households are probably much more likely to also have children to support.

Education has a fair positive effect, likely as it means greater opportunities and better financial decision making.

The effect size for 'age of household head' is small enough that it lacks practical significance despite being statistically significant.

Figure 5

Relationship between food expenditure as a percentage of household expenditure and the log of household's daily expenditure per capita in Tanzania



The scatterplot above shows an obvious slight downward trend between the percentage of household expenditure spent on food and the log of a household's daily expenditure per capita, suggesting Engel's law would hold true despite the large spread. The line of best fit in figure 5 was drawn using a linear model and was found to be statistically significant. Using a Pearson's product-moment correlation, the correlation was found to be -0.486, meaning there is a moderate negative correlation between percent food expenditure and log daily per-capita expenditure. Therefore, Engel's law holds roughly true in Tanzania despite the large amount of spread in household level data.